

Cryptographic Hash Functions

CS 171: Cryptography

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October 6, 2026

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Hash Functions

Cryptographic hash function

A **deterministic** function mapping an arbitrarily long input string to a shorter, fixed-length output string.

- Hash functions can be **keyed** or **unkeyed**.
 - **In theory**: keyed.
 - **In practice**: unkeyed (fix a key once and for all).

Hash Function Definition

Hash function and collisions

A hash function is $H : \{0, 1\}^* \rightarrow \{0, 1\}^\ell$.

A **collision** is a pair of *distinct* inputs x and x' such that $H(x) = H(x')$.

- Classical use is in **data structures**, where collisions are *undesirable*.
- For cryptographic hash functions, avoiding collisions is a **requirement** — even when an attacker is *maliciously* trying to find them.

Hash Function Definition (Keyed)

Definition 1 (Hash function)

A hash function with output length ℓ is a pair of algorithms (Gen, H) :

- $\text{Gen}(1^n)$ (PPT): outputs a key s .
- H (deterministic, polynomial time): on input a key s and a string $x \in \{0, 1\}^*$, outputs $H_s(x) \in \{0, 1\}^{\ell(n)}$.

Fixed-length hash function

If H_s is defined only for inputs $x \in \{0, 1\}^{\ell'(n)}$ with $\ell'(n) > \ell(n)$, then (Gen, H) is a **fixed-length** hash function for inputs of length ℓ' : it **compresses**.

- The key s is **public**: it is not a secret key.

Collision Resistance

Game $\text{Hash-coll}_{\mathcal{A}, \Pi}(n)$

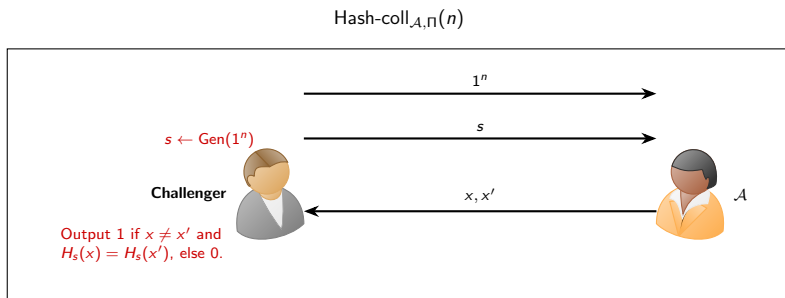
- 1 $s \leftarrow \text{Gen}(1^n)$. $\mathcal{A}$ is given 1^n and the key s .
- 2 $\mathcal{A}$ outputs a pair x, x' .
- 3 Output 1 if $x \neq x'$ and $H_s(x) = H_s(x')$, and 0 otherwise.

Definition 2 (Collision resistance)

$\Pi = (\text{Gen}, H)$ is **collision resistant** if for all PPT $\mathcal{A}$:

$$\Pr [\text{Hash-coll}_{\mathcal{A}, \Pi}(n) = 1] \leq \text{negl}(n).$$

Hash-coll_{A,Π}: Challenger vs. Adversary



No secrets!

$\mathcal{A}$ gets the key s , and there are no oracles: the game has nothing to hide. A collision always *exists* (the function compresses); it must be hard to *find*.

Hash Functions in Practice

- Have a **fixed output length**, just like block ciphers.
- Also, they are **unkeyed**: a single function is fixed once and for all.
 - **Problematic in theory**: for a fixed function, a collision x, x' exists, so the adversary that simply outputs it (“hard-wired”) wins with probability 1.
 - In practice, nobody knows how to write that adversary down.

Generic Attacks on Hash Functions

Setting

Fixed-length hash function $H : \{0, 1\}^{\ell'} \rightarrow \{0, 1\}^{\ell}$ with $\ell' > \ell$. A collision is distinct x, x' with $H(x) = H(x')$.

- We drop the key subscript s , which is now implicit.

Can we find collisions?

Yes. Let $x_1, \dots, x_{2^\ell+1}$ be arbitrary distinct values in $\{0, 1\}^{\ell'}$. There are only 2^ℓ outputs, so by pigeonhole

$$\exists i \neq j : H(x_i) = H(x_j).$$

But this takes $2^\ell + 1$ evaluations of H .

Can We Find Collisions Faster?

Setting

$H : \{0,1\}^{\ell'} \rightarrow \{0,1\}^{\ell}, \ell' > \ell.$

- Let $x_1, \dots, x_q$ be distinct values in $\{0,1\}^{\ell'}$. What is the probability that two of them collide?
- When $q > 2^{\ell}$, the probability is 1. What if q is smaller?

Important

A much smaller q suffices:

$$q \approx 2^{\ell/2}.$$

Birthday attack: hash q distinct inputs and look for a repeated output (e.g. by sorting the q values).

Heuristic Analysis: The Birthday Bound

View H as a random function

For distinct $x_1, \dots, x_q$, the values $H(x_1), \dots, H(x_q)$ are then independent and uniform in $\{0, 1\}^\ell$, and

$$\Pr[\exists i \neq j : H(x_i) = H(x_j)] \approx 1 - e^{-q(q-1)/(2 \cdot 2^\ell)} \approx \frac{q^2}{2 \cdot 2^\ell} \quad (\text{for small } q).$$

- Thus the probability is $\frac{1}{2}$ for $q = \Theta(2^{\ell/2})$.

Where Does $q^2/2^{\ell+1}$ Come From?

Write $N = 2^\ell$ and let Coll be the event that $H(x_i) = H(x_j)$ for some $i \neq j$.

Attempt 1

Each pair collides with probability $1/N$, and there are $\binom{q}{2}$ pairs, so $\Pr[\text{Coll}] = \binom{q}{2} \cdot \frac{1}{N}$?

Error: the pair events are **not independent** (nor disjoint), so the sum over-counts. It is only an **upper bound** (union bound): $\Pr[\text{Coll}] \leq \frac{q(q-1)}{2N}$.

Lower bound: no collision means each new value avoids all earlier ones

$$\begin{aligned}\Pr[\neg \text{Coll}] &= \prod_{i=1}^{q-1} \left(1 - \frac{i}{N}\right) && (H(x_{i+1}) \text{ avoids } i \text{ distinct values}) \\ &\leq \prod_{i=1}^{q-1} e^{-i/N} && (1 - z \leq e^{-z}) \\ &= e^{-q(q-1)/2N}.\end{aligned}$$

So $\Pr[\text{Coll}] \geq 1 - e^{-q(q-1)/2N}$, which is about 0.63 already for $q \approx \sqrt{2N} = 2^{(\ell+1)/2}$.

The Birthday Problem

With q people, what is the probability that two share a birthday? ($N = 365$.)

- By the formula of the previous slide, $\Pr[\text{Coll}] \geq \frac{1}{2}$ once

$$1 - e^{-q(q-1)/(2 \cdot 365)} \geq \frac{1}{2} \iff q(q-1) \geq 2 \ln 2 \cdot 365 \approx 506.$$

- $q = 23$ is the first such q : $23 \cdot 22 = 506$. (Exactly: $\Pr[\text{Coll}] \approx 0.507$ for 23 people, and only ≈ 0.41 for 20.)
- In general $q \approx \sqrt{2 \ln 2 \cdot N} \approx 1.18\sqrt{N}$: $\Theta(\sqrt{N})$, as for hash outputs with $N = 2^\ell$.

Implications of the Birthday Attack

Required output length

To resist attackers running in time 2^n , we need hash output length

$$\ell = 2n.$$

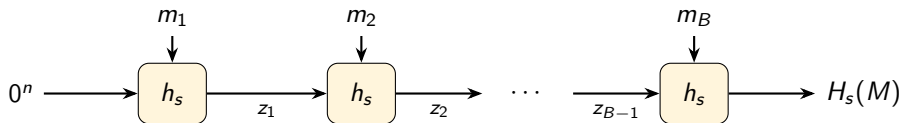
- This is **double** the key length needed for block ciphers.
- So to get 128-bit security we need a hash output of 256 bits.
- **Necessary but not sufficient**: the birthday attack works against *all* hash functions, but a specific function may have other, more “devastating” attacks.

Domain Extension: Merkle–Damgård (First Attempt)

Construction

Given (Gen, h) , a fixed-length hash function from $2n$ -bit inputs to n -bit outputs, construct (Gen, H) :

$H_s(M)$: parse M as n -bit blocks $m_1 \cdots m_B$, where m_B is padded with 0s to make it of the appropriate length. Let $z_0 = 0^n$, $z_i = h_s(z_{i-1} \parallel m_i)$, and output z_B .



Is it secure?

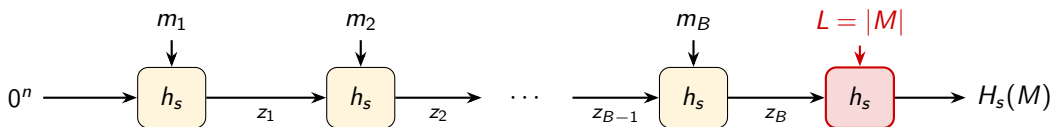
No! If $|M|$ is not a multiple of n , then M and $M \parallel 0$ are padded to the *same* blocks: $H_s(M) = H_s(M \parallel 0)$, a collision.

Domain Extension: The Merkle–Damgård Transform

Construction (fixed)

As before, but append one more block $m_{B+1} := L$, the length $L = |M|$ encoded in n bits:

$$H_s(M) := z_{B+1} = h_s(z_B \parallel L).$$



- M and $M\parallel 0$ now have different lengths, so their last blocks differ.
- Messages of any length $< 2^n$ can be hashed, in one pass over M .

Theorem 3

If (Gen, h) is collision resistant, then so is (Gen, H) .

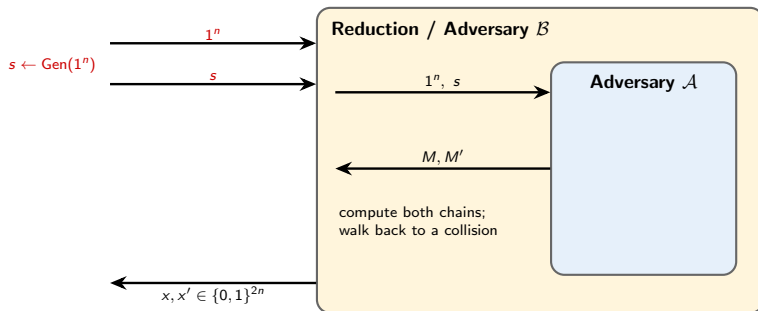
Proof by reduction

Given a PPT adversary $\mathcal{A}$ that finds collisions in H , construct a PPT adversary $\mathcal{B}$ that finds collisions in h , **with the same key** s .

Idea

A collision $M \neq M'$ on H_s is two chains of h_s ending in the same value. Walk back along them from the end: somewhere, h_s is applied to two **different** inputs and gives the **same** output.

Merkle–Damgård: Proof by Reduction

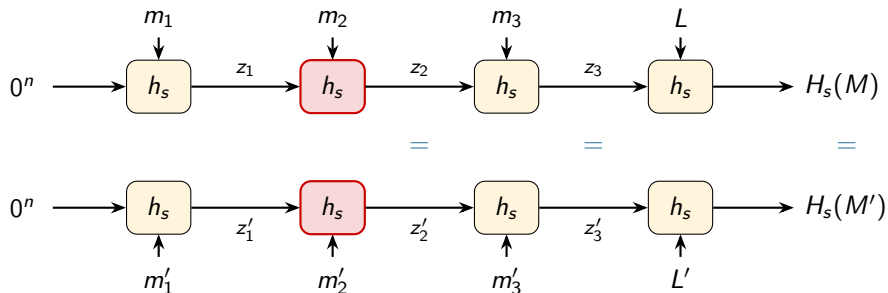


Given / To prove

Given: $\Pr[\text{Hash-coll}_{\mathcal{A}, (\text{Gen}, H)}(n) = 1] \geq \epsilon(n)$ for non-negligible ϵ .

To prove: $\Pr[\text{Hash-coll}_{\mathcal{B}, (\text{Gen}, h)}(n) = 1] \geq \epsilon(n)$.

Walking Back Along the Chains



- 1 The outputs are equal: $H_s(M) = H_s(M')$.
- 2 Inputs to the last h_s : $z_3 || L = z'_3 || L'$. Otherwise they are a collision; since they are equal, $z_3 = z'_3$.
- 3 Inputs to the third h_s : $z_2 || m_3 = z'_2 || m'_3$ (here $m_3 = m'_3$), so $z_2 = z'_2$.
- 4 Inputs to the second h_s : $z_1 || m_2 \neq z'_1 || m'_2$, but the outputs $z_2 = z'_2$: **a collision on h_s !**

Finding the Collision: Two Cases

Write $l_i = z_{i-1} \parallel m_i$ for the input to the i -th call of h_s , with $m_{B+1} = L$. The outputs agree: $z_{B+1} = z'_{B'+1}$.

Case $|M| \neq |M'|$

The last calls are a collision: $(l_{B+1}, l'_{B'+1})$.

$l_{B+1} = z_B \parallel L$ and $l'_{B'+1} = z'_{B'} \parallel L'$ differ, since $L \neq L'$, and both give $h_s(\cdot) = H_s(M) = H_s(M')$.

Case $|M| = |M'|$ (so $B = B'$)

Let i^* be the **largest** i with $l_i \neq l'_i$. Then (l_{i^*}, l'_{i^*}) is a collision.

i^* exists: $M \neq M'$ have the same length, so some block $m_i \neq m'_i$.

If $i^* = B + 1$: equal outputs. Else

$l_{i^*+1} = l'_{i^*+1}$, so its first half gives $z_{i^*} = z'_{i^*}$, i.e. $h_s(l_{i^*}) = h_s(l'_{i^*})$.

Merkle–Damgård: Putting It Together

Success probability of $\mathcal{B}$

$\mathcal{B}$ gives $\mathcal{A}$ exactly the input of $\text{Hash-coll}_{\mathcal{A},(\text{Gen},H)}(n)$, and whenever $\mathcal{A}$ outputs a collision on H_s , $\mathcal{B}$ outputs a collision on h_s (two cases). So

$$\Pr[\text{Hash-coll}_{\mathcal{B},(\text{Gen},h)}(n) = 1] \geq \Pr[\text{Hash-coll}_{\mathcal{A},(\text{Gen},H)}(n) = 1] \geq \epsilon(n),$$

contradicting that (Gen, h) is collision resistant. □

Remark (running time)

$\mathcal{B}$ runs $\mathcal{A}$ once and then evaluates h_s along both chains: $B + B' + 2$ calls, polynomial since $\mathcal{A}$ wrote M and M' in polynomial time.

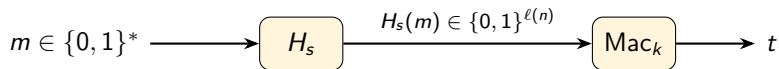
Takeaway

No probabilities to juggle: every collision on H_s *contains* a collision on h_s .

MACs using Hash Functions: Hash-and-MAC

- Previously: constructions of MACs from a PRF / block cipher.
- **CBC-MAC** gave MACs with short tags for arbitrary-length messages.
- The **Hash-and-MAC** paradigm does the same with a hash function: hash the long message to $\ell(n)$ bits, then MAC the hash.

Hash-and-MAC



Construction

Let $\Pi = (\text{Gen}, \text{Mac}, \text{Vrfy})$ be a MAC for messages of length $\ell(n)$, and let $\Pi_H = (\text{Gen}_H, H)$ be a hash function with output length $\ell(n)$. The MAC $\Pi' = (\text{Gen}', \text{Mac}', \text{Vrfy}')$ for arbitrary-length messages:

- $\text{Gen}'(1^n)$: output $k' = (k, s)$, where $k \leftarrow \text{Gen}(1^n)$ and $s \leftarrow \text{Gen}_H(1^n)$.
- $\text{Mac}'_{k'}(m)$, for $m \in \{0, 1\}^*$: output $t \leftarrow \text{Mac}_k(H_s(m))$.
- $\text{Vrfy}'_{k'}(m, t)$: output 1 iff $\text{Vrfy}_k(H_s(m), t) = 1$.

Hash-and-MAC: Security

Theorem 4

If Π is a secure (eu-cma) MAC for messages of length $\ell(n)$ and Π_H is collision resistant, then Π' is a secure MAC for arbitrary-length messages.

Proof sketch

Say $\mathcal{A}$ queries $m_1, \dots, m_q$ and outputs a forgery (m^*, t^*) , $m^* \notin \{m_1, \dots, m_q\}$.

- **Case I:** $H_s(m^*) = H_s(m_i)$ for some i — then (m^*, m_i) is a **collision** on H_s .
- **Case II:** $H_s(m^*) \neq H_s(m_i)$ for all i — then $(H_s(m^*), t^*)$ is a **forgery** for Π : a valid tag on a hash value whose tag was never requested.

Next: each case is a reduction.

Hash-and-MAC: Two Kinds of Forgeries

Let win be the event $\text{MacForge}_{\mathcal{A}, \Pi'}(n) = 1$, and Coll the event that $H_s(m^*) = H_s(m_i)$ for some query m_i of $\mathcal{A}$. Then

$$\Pr[\text{win}] = \Pr[\text{win} \wedge \text{Coll}] + \Pr[\text{win} \wedge \neg \text{Coll}].$$

Case Coll: a collision

$\mathcal{B}_1$ breaks collision resistance of Π_H :

$$\Pr[\text{Hash-coll}_{\mathcal{B}_1, \Pi_H}(n) = 1] \geq \Pr[\text{win} \wedge \text{Coll}].$$

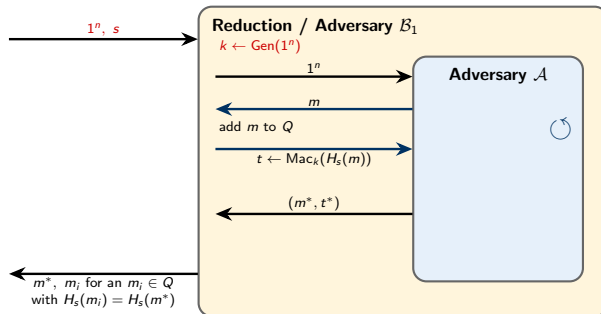
Case $\neg \text{Coll}$: a forgery on a new hash

$\mathcal{B}_2$ breaks unforgeability of Π :

$$\Pr[\text{MacForge}_{\mathcal{B}_2, \Pi}(n) = 1] \geq \Pr[\text{win} \wedge \neg \text{Coll}].$$

Both reductions run $\mathcal{A}$ and answer its queries exactly as $\text{Mac}'_{k'}$ would; they differ in which key they are *given* and which they *sample*.

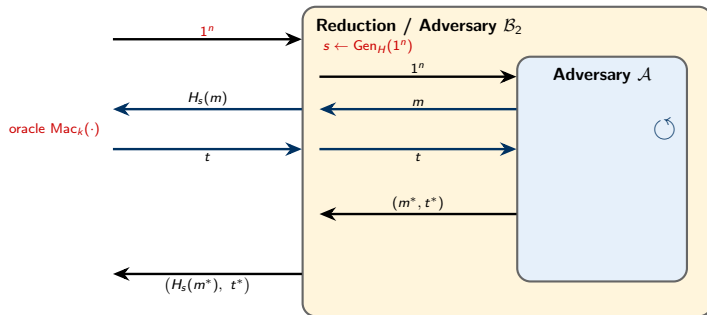
Case Coll: The Reduction $\mathcal{B}_1$



Claim

$\mathcal{B}_1$ holds k and s , so $\mathcal{A}$'s view is exactly as in $\text{MacForge}_{\mathcal{A}, \Pi'}(n)$. On $\text{win} \wedge \text{Coll}$: $m^* \notin Q$, so $m^* \neq m_i$, and $H_s(m^*) = H_s(m_i)$ — a collision.

Case $\neg\text{Coll}$: The Reduction $\mathcal{B}_2$



Claim

Each answer is $\text{Mac}_k(H_s(m)) = \text{Mac}'_{k,s}(m)$, so $\mathcal{A}$'s view is exactly as in $\text{MacForge}_{\mathcal{A},\Pi'}(n)$. On win $\wedge \neg\text{Coll}$: $\text{Vrfy}_k(H_s(m^*), t^*) = 1$ and $\mathcal{B}_2$ never queried $H_s(m^*)$ — a forgery.

Hash-and-MAC: Putting It Together

Success probability of $\mathcal{B}_1$ and $\mathcal{B}_2$

Suppose $\Pr [\text{MacForge}_{\mathcal{A}, \Pi'}(n) = 1] \geq \epsilon(n)$ for non-negligible ϵ . Then

$$\begin{aligned}\epsilon(n) &\leq \Pr[\text{win} \wedge \text{Coll}] + \Pr[\text{win} \wedge \neg \text{Coll}] \\ &\leq \Pr [\text{Hash-coll}_{\mathcal{B}_1, \Pi_H}(n) = 1] + \Pr [\text{MacForge}_{\mathcal{B}_2, \Pi}(n) = 1].\end{aligned}$$

So for each n , at least one of the two terms is $\geq \epsilon(n)/2$: $\mathcal{B}_1$ finds collisions or $\mathcal{B}_2$ forges with non-negligible probability — contradicting that Π_H is collision resistant or that Π is secure. □

Remark (running time)

$\mathcal{B}_1$ and $\mathcal{B}_2$ run $\mathcal{A}$ once, plus one hash and one tag (or one oracle query) per query: PPT.

Other Applications of Hash Functions

- Blockchains.
- Virus fingerprinting.
- Deduplication.
- Peer-to-peer (P2P) file sharing.

Summary

- **Cryptographic hash functions** (Gen, H) compress long inputs to $\ell(n)$ bits; security is **collision resistance**, via the game $\text{Hash-coll}_{\mathcal{A}, \Pi}(n)$, in which the key s is public.
- **Birthday attack**: $q \approx 2^{\ell/2}$ evaluations find a collision in *any* hash function, so the output length must be $\ell = 2n$ for 2^n security.
- **Merkle–Damgård** extends a compressing h to any length; with the length block, collision resistance is preserved. *Proved*: walking back along the two chains finds a collision on h .
- **Hash-and-MAC**: $\text{Mac}_k(H_s(m))$ is a secure MAC for arbitrary-length messages. *Proved*: a forgery is either a collision on H_s ($\mathcal{B}_1$) or a forgery on a new hash value ($\mathcal{B}_2$).